

Anti-Soviet spy bug

In 1982, the CIA allegedly planted a bug in the Canadian computer system, which the Soviet government had purchased to manage its Siberian pipeline

AT&T hit by the bug

In 1990, AT&T released software infected with a bug that caused many of the company's mammoth computers to crash causing network outage

Feature

huge R&D facility in different parts of the country. It won't be wrong to say that the backbone of these facilities is made out of the Software Quality Assurance departments. In 2005, Adobe had been ranked 15th on the list of best places to work for in India. Viraj, who is employed in the Software QA department at the Noida headquarters of Adobe India, says, "The work atmosphere is simply the best. And that's what makes the atmosphere so charged and great fun." In fact, the field of Software Quality Assurance is so vast that it corners a sizable percentage of the total operating budget in almost all the top IT companies in the world.

Highly paying jobs

As far as the salaries go, Software Quality Assurance is considered to be one of the most important departments in any technical company, and hence the professionals employed in such departments are highly paid. A fresher can expect a salary ranging between Rupees 25,000 to Rupees 50,000. The size of the paycheck will depend on the size and scope of the company. Also, students with a B.E./B.Tech degree and a latest certification are paid much more than those with only a diploma in engineering or science. Those with MCA or Masters degrees in computer science might be able to draw the highest salary. However, once an individual has joined the company, further progress is usually linked to his or her

performance. The designations for this line of work differ from company to company. The most common designations include - Associate Software Engineer, Software Verification Engineer, Technical manager, etc.

A study done by NIST (National Institute of Standards and Technology) in USA has come to the conclusion that in USA alone almost a hundred billion dollars are lost due to bugs in software. The loss figures for the entire world would be much larger. Software companies have realised that much of this money can be saved if better testing and quality assurance practices are implemented. So the importance of the Quality Assurance department is well recognised all over the world. If you choose this line of career than within a period of five to ten years you can expect a significant advancement, in terms of perks and privileges. A Software QA engineer with an experience of 5 to 10 years can easily expect a monthly compensation of more than Rupees 75000. Of course, the final compensation must vary from company to company, and it will also depend on the performance of the engineer.

Taste of Software

There are many different types of testing that are currently in use, but broadly these systems can be divided into two categories - White Box testing and Black Box testing. In case of White Box testing, the engineers take an internal view of the software, whereas in Black Box testing, an external view is taken. The testing can also be static or dynamic. Merely reviews or inspections are considered to be examples of static testing. In dynamic testing, the program is actually executed so that its operational efficiency in the live environment can be analysed. Over the years, many new software tools have been developed to make the job of testing new software easy. However, even with better tools we are still far from the stage where software performance can be perfect. According to Capers Jones,

founder of Software Productivity Research (SPR), "The number of bugs in newly written code has remained constant at around five per function point." Jones further states that only 85% of bugs are usually removed before software is put into use.

One problem that professionals face is that most firms no longer have all their programmers and testers in the same place, or for that matter in the same country. So it is possible that one campus might be testing or assessing the quality of only one segment of the software code, while a campus located elsewhere might be handling the other segment. It is necessary for the engineers to be able to communicate with each other in real time, so that they can request design changes, report bugs and basically coordinate their efforts. Engineers need to be able to decide which bugs are really worth fixing. If the bug is not doing any harm to the operations of the software, then the best option might be to ignore it and move on to solving the really big problems. Along with knowledge of coding, the field of Software Quality Assurance also requires the ability of stepping into the shoes of the actual end user, so that you can focus on the kind of product that will actually work for him.

While shedding light on the job that he actually performs, Sanjay Prakash says, "I do not carry with me the feeling that all software has bugs, rather my job is to let my company know that this is the amount of time and money it is going to take for making a software of this quality. Basically you have to take the budget into account in order to decide what quality of software you are going to create." The bottom line is that software testing is an art, in the sense that good testing also requires an engineer's creativity, experience and intuition, together with proper techniques. There is no doubt that the challenges that Software Quality Assurance engineer faces are stiff, but if you can manage to take all the hard work in your stride, then you can make a comfortable living. 

Major software certifications by QAI

Associate-Level Certifications

- Certified Associate in Software Quality (CASQ)
- Certified Associate in Software Testing (CAST)

Professional Certifications

- Certified Software Quality Analyst (CSQA)
- Certified Software Tester (CSTE)
- Certified Software Project Manager (CSPM)
- Certified Software Business Analyst (CSBA)
- Certified Software Process Engineer (CSPE)
- Certified Quantitative Software Process Engineer (CQSPE)

Advanced Certifications (Professional Certification is a prerequisite)

- Certified Manager of Software Testing (CMST)
- Certified Manager of Software Quality (CMSQ)